

Exercise13

CPS I — Exercise Sheet 13

17.75

University of Freiburg · Winter Term 2025/26

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Exercise 1: LTL Properties

(a)satisfies

- $\{a\{b^\omega\}$ ✓
- $\{a\{a\{b^\omega\}$ ✓
- $\{c\{b^\omega\}$ ✓
- $\{c\{a^\omega\}$ ✓
- $\{a^\omega\}$ ✓
- $\{c^\omega\}$ ✓

(b)violates

- $\{a^\omega\}$ ✓
 - $\{a\{c^\omega\}$ ✓
 - $\{a\{b^\omega\}$ ✓
 - $\{a^\omega\}$ ✓
 - $\{b^\omega\}$ ✓
 - $\{c\{a^\omega\}$ ✓
- $\{a\}^\omega$ satisfies ϕ_4 - 0.5

(c)TS

- No, because $A_0 = \{b$ or $A_0 = \{a, c$ $A_1 = \{a$ dont satisfy a and next b ✓
- Yes, $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$? - 0.25
- No, because $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$ satisfies $aU \Box b$
- NO • Yes, $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$ - 0.5
- Yes, $s_1 \rightarrow s_2 \rightarrow (s_3 \rightarrow)^\omega$ - 0.25
- Yes, infinite c. $(s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow)^\omega$ NO - 0.5

If yes: explain why all traces of the TS satisfy ϕ_i
If no: give counter example trace that does not satisfy ϕ_i

(d)set comprehension

- $Words(\phi_1) = \{A_0A_1... \in (2^{AP})^\omega | a \in A_0 \wedge b \in A_1\}$ ✓
- $Words(\phi_2) = \{A_0A_1... \in (2^{AP})^\omega | \exists k \in N. (b \in A_k \wedge \forall i < k. a \in A_i)\}$ ✓
- $Words(\phi_3) = \{A_0A_1... \in (2^{AP})^\omega | \neg(\exists k \in N. (\forall j \geq k. b \in A_j) \wedge (\forall i < k. a \in A_i))\}$ ✓

- $Words(\phi_4) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \exists k \in N. (c \in A_k \wedge \forall i > k. a \in A_i)\}$ — 0.5
- $Words(\phi_5) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \exists k \in N. \forall i \geq k. a \in A_i\}$ ✓
- $Words(\phi_6) = \{A_0A_1\ldots \in (2^{AP})^\omega \mid \forall k \in N. \exists i \geq k. a \in A_i\}$ (✓) ✓

Exercise 2: Stating properties in LTL 4.5/4 + 1 Bonus

- (a) $\Box \neg (\text{Peter.use} \wedge \text{Betsy.use})$ ✓
- (b) $\Box (\text{Peter.use} \rightarrow \Diamond \text{Peter.release}) \wedge \Box (\text{Betsy.use} \rightarrow \Diamond \text{Betsy.release})$ ✓
- (c) $\Box (\text{Peter.request} \rightarrow \Diamond \text{Peter.use}) \wedge \Box (\text{Betsy.request} \rightarrow \Diamond \text{Betsy.use})$ ✓
- (d) $\Box (\text{Peter.request} \rightarrow \Diamond \neg \text{Peter.request}) \wedge \Box (\text{Betsy.request} \rightarrow \Diamond \neg \text{Betsy.request})$ ✓
- (e) $\Box ((\text{Peter.use} \rightarrow (\neg \text{Peter.use} \cup \text{Betsy.use})) \wedge (\text{Betsy.use} \rightarrow (\neg \text{Betsy.use} \cup \text{Peter.use})))$
 P.use U B.use U — 0.5

Exercise 3: From Set Notation to LTL

- (a) $\phi_1 = \Diamond (a \wedge \bigcirc b)$ ✓
- (b) No ✓
- (c) $\phi_3 = \Box ((a \leftrightarrow \bigcirc \bigcirc a) \wedge (b \leftrightarrow \bigcirc \bigcirc b))$ ✓
- (d) $\phi_4 = \Box (a \rightarrow \bigcirc b)$ — 0.5

3.5/4